Agarose salt bridges

March 16, 2020.

Required materials

- Low melting agarose.
- Salt solution (3M KCl).
- Box for the agarose bridges, filled on the bottom with 3M KCl solution.
- Yellow pipet tips (20-200 µL).
- PCR-tube caps (or other fitting caps).

Method

Making the agarose bridges

1. Dissolve 100 mg low melting agarose in 3.35 mL 3M KCl solution (3 mass% agarose).
2. Heat the agarose/salt solution to ~70°C such that the agarose dissolves.
3. Mix the agarose/salt solution (at this step, do not cool down below ~40°C).
4. Add 75 µL agarose/salt solution to the (yellow) pipet tips, such that the tip contains agarose (make sure there are no air bubbles in the agarose).
5. Directly put the agarose bridges in the box with 3M KCl solution, such that the tip touches the salt solution.
6. Completely fill the top part of the agarose bridges with 3M KCl solution.

Checking the agarose bridges

1. Check if the salt bridges were sealed correctly (loss of 3M KCl solution from top part of the agarose bridges means leaky agarose bridge).
2. Cap the agarose bridges that are not leaking and throw away the leaky agarose bridges.
    - Take PCR-tube caps and puncture a hole in the middle (to fit the electrode).
    - Caps can be reused between bridges, so there is no need to cap all bridges at once.
    - Salt will accumulate on the cap over time, clean the cap and add fresh 3M KCl solution.
3. Store the agarose salt bridges at 4°C for up to 90 days.

Note:
Agarose bridges may get infected with fungi, so check them every time before use.

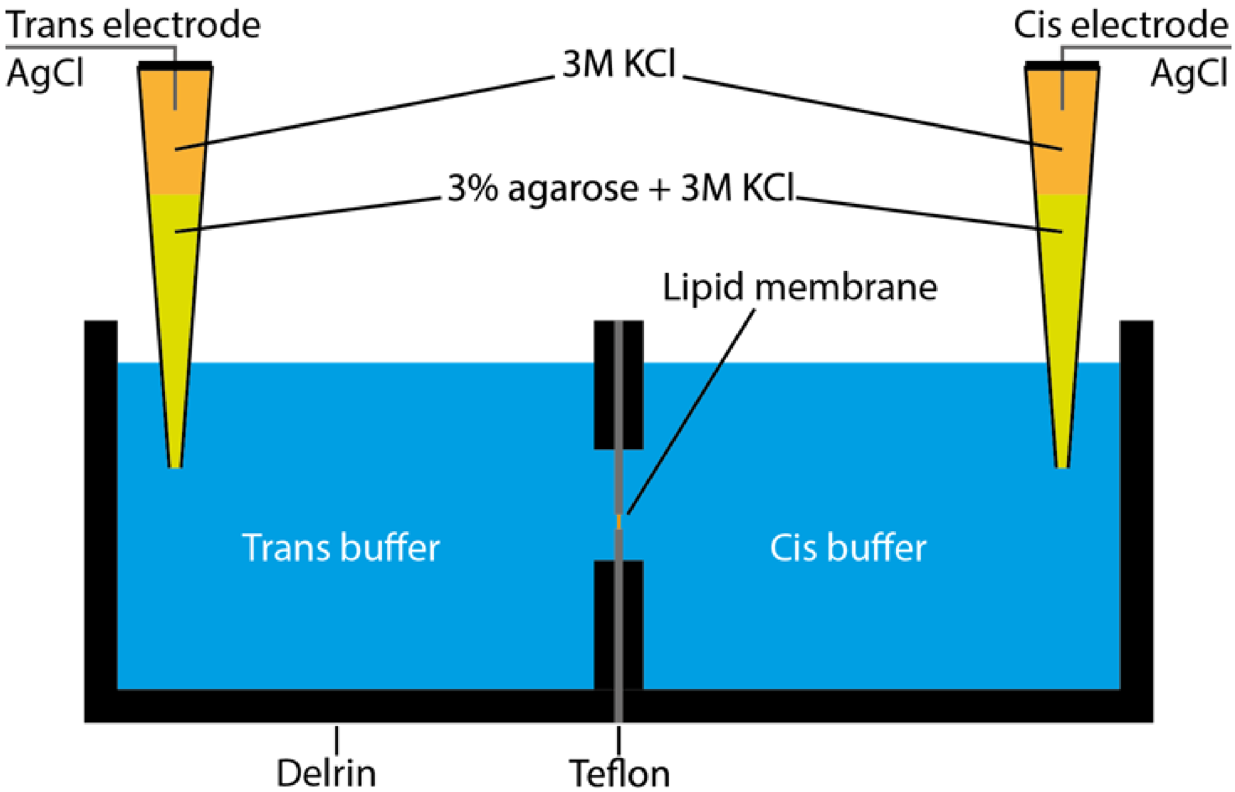

Figure 1: Illustration of the use of agarose bridges inside a nanopore chamber. The electrodes are dipped inside the top compartment of the agarose bridges and connected to the surface of the nanopore chamber.